

# Basic Details of the Team and Problem Statement



**Problem Statement Number : P11**

**Problem Statement Title :** AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant for Rural Micro-Entrepreneurs

**Project Name :** GramSaarthi AI (ग्रामसारथी)

**Team Name :** Rural Innovators / GramSaarthi Team

**Team Leader Name :** Chirag Vasava

**GitHub Repository :**

<https://github.com/ChiragVasava/gramsaarthi-ai.git>

# Idea/Approach Details

## Describe your Idea/Solution/Prototype here:

- **Core Problem Addressed:** 70M+ rural micro-entrepreneurs face a 40%+ failure rate due to informal hearsay business planning and severely underutilized concessional credit schemes.
- **GramSaarthi AI Solution:** An intelligent, trilingual rural advisory platform combining two decoupled engines:
  - **1. Hyper-Local Feasibility Engine:** Analyzes 5 km & 10 km spatial catchment zones, demographic demand clustering, competitor density, seasonal threats, and live commodity pricing.
  - **2. Statutory Financial Structuring Engine:** Implements audit-grade statutory debt sizing (10% Margin, 90% Debt), automatic scheme routing (Micro Finance vs. Term Loan), moratorium grace modeling, and quarterly amortization.
- **Institutional Feasibility Dossier:** Generates instant bank-grade loan appraisal dossiers exportable as PDF for formal credit underwriting.

## Describe your Technology stack here:

- **Frontend & App Router:** Next.js 16 (React 19) with lightning-fast static page generation & server rendering.
- **Language:** TypeScript 5 for strict mathematical type-safety across financial engines.
- **Styling & UI:** Tailwind CSS 4 + Lucide Icons + print stylesheets for PDF dossiers.
- **Database & ORM:** SQLite + Prisma ORM for zero-setup, self-contained evaluation.
- **Spatial Mapping:** Leaflet.js radial GIS radar (5 km & 10 km catchment analytics).
- **AI & Fallback Tier:** Google Gemini API + Deterministic Fallback Engine (100% demo uptime)
- **Voice & Inclusivity:** Native Web Speech API supporting English, Hindi (HI), and Gujarati (GU).

# Idea/Approach Details

## Describe your Use Cases here

### ► Dairy / Livestock Unit Setup (e.g. 5-Cow Dairy):

- Founder enters ₹50,000 margin capital in 4-step wizard.
- System automatically sizes ₹5,00,000 project cost and ₹4,50,000 debt under Term Loan Scheme (8.0% p.a., 7 yrs, 6-mo moratorium).
- Spatial radar evaluates 5 km milk collection hubs, Amul chilling centers, and veterinary clinics.

### ► Rural Kirana / Agro-Retail Expansion:

- Sizes ₹1,40,000 project with ₹14,000 margin under Micro Finance Scheme (6.5% p.a., 3 yrs, 3-mo moratorium).
- Calculates working capital buffer for seasonal inventory.

### ► Institutional Loan Application Preparation:

- Generates audit-ready Feasibility Dossier with SWOT analysis and quarterly repayment schedule ready for Bank/SCA submission.

### ► Voice-First Trilingual Advisory for Bharat:

- Rural founders ask questions via voice in Hindi or Gujarati and receive contextual guidance on loan repayments and threat mitigation.

## Describe your Dependencies

### ► Core Software Dependencies:

- Node.js 18+ runtime environment
- Next.js 16.3 / React 19 web framework
- Prisma 5.22 ORM with SQLite database
- Tailwind CSS v4 & Lucide React icons

### ► External Services & APIs:

- Google Gemini API (Generative advisory & SWOT)
- OpenStreetMap / Carto CDN (Tile layer rendering)
- Web Speech Synthesis & Speech Recognition API

### ► Hardware & Deployment Prerequisites:

- Runs on any standard PC, laptop, or modern mobile browser
- Zero cloud infrastructure required: 100% portable local SQLite database with demo credentials pre-seeded
- Offline resilience: Autonomous deterministic fallback ensures zero API quota lockouts during live presentations

# Idea/Approach Details

## Uniqueness of your Idea

- **★ Decoupled Deterministic Financial Engine:** Unlike naive LLM wrappers that hallucinate loan math, debt sizing, interest rates, and amortizations are computed by an exact deterministic TypeScript algorithm conforming to statutory guidelines.
- **★ Automated Statutory Scheme Routing:** Intelligently routes projects  $\leq ₹1.40L$  to Micro Finance (6.5%, 3-yr, 3-mo moratorium) and  $> ₹1.40L$  to Term Loan (8.0%, 7-yr, 6-mo moratorium) without manual loan officer intervention.
- **★ Hyper-Local Concentric Catchment (5 & 10 km):** Provides granular spatial demographic density, competitor proximity, and mandi linkage rather than generic national advice.
- **★ Trilingual Rural Accessibility:** Voice input and localized UI in English, Hindi, and Gujarati overcomes rural literacy barriers.
- **★ Production-Ready & Fully Functional:** Not just mockups—a fully implemented web app with 10 working routes, SQLite database, and instant PDF dossier generation.

## Future Scope and Impact

- **► Measurable Socio-Economic Impact:**
  - Reduces 40%+ early enterprise mortality by replacing hearsay with data-driven demand validation.
  - Unlocks statutory concessional debt for marginalized rural founders, replacing predatory moneylenders (36%+ p.a.) with 6.5%–8% subsidized credit.
  - Cuts bank dossier appraisal preparation time from 3 weeks to under 3 minutes.
- **► Future Engineering Scope & Roadmap:**
  - SCA & JanSamarth Portal API: Direct automated digital loan application dispatch to State Channelizing Agencies & national banks.
  - Live Mandi & APMC Feeds: Real-time price tracking via e-NAM APIs for hyper-local commodity arbitrage guidance.
  - Offline PWA & Edge AI: Local on-device LLM execution (Gemma 2B / WebLLM) for zero-connectivity deep rural villages.

# Architecture Diagram

## GramSaarthi AI — End-to-End System Architecture

Decoupled Architecture (P11)

### 1. Presentation Layer (Next.js 16)

#### 4-Step Guided Feasibility Wizard

/wizard

Step-by-step intake: Geography, Trade, Capital presets, & Founder

#### Hyper-Local Market Radar Map

/map

Interactive Leaflet radar map with 5 km & 10 km concentric catchm

#### Statutory Financial Calculator

/calculator

Live statutory sizing, scheme selector, moratorium, & quarterly am

#### Trilingual AI Advisory Desk

/chat

Contextual voice & text assistant with English, Hindi (HI), and Gujar

#### Institutional Feasibility Dossier

/report

Complete bank-grade appraisal report with instant print-to-PDF ex

### 2. Deterministic Financial Micro-Engine

#### Statutory Sizing & Scheme Router (Audit-Grade)

- Project Cost Multiplier:  $\text{Project Cost} = \text{Margin Capital} / 0.10$  (10%)
- Concessional Debt Sizing:  $\text{Loan Amount} = \text{Project Cost} * 0.90$  (90%)
- **Statutory Routing:**  $\text{Project} \leq 1.40L \rightarrow \text{Micro Finance Scheme (8.0\%)}$   
 $\text{Project} > 1.40L \rightarrow \text{Term Loan Scheme (8.0\%)}$

#### Moratorium & Amortization Schedule

- 3-Month (Micro) or 6-Month (Term Loan) Interest-Only Moratorium
- Dynamic quarterly principal, interest & residual balance schedules.
- 3-Month Working Capital Reserve calculated per industry trade.

#### Spatial Demographics & Catchment Analytics

- Primary Zone (5 km): Immediate customer cluster, footfall, weekly h
- Secondary Zone (10 km): Institutional buyers, APMC mandis, dairy h
- Competitor Density Index & Local Threat Warning System.

#### Multilingual NLP & Speech Normalization

- In-browser Web Speech API for low-bandwidth rural voice queries.
- Prompt Context Injection with current project state & balance sheet
- Trilingual translation & transliteration support (EN, HI, GU).

### 3. Intelligence & Persistence Tier

#### Gemini 1.5 Flash Reasoning API

Qualitative Advisory

Generates SWOT analysis, competitive risk mitigation, and commodity p

#### Deterministic Fallback Engine

Zero-Failure Guarantee

Autonomous offline fallback generator ensuring 100% operational upti

#### SQLite Database + Prisma ORM

Local Data Persistence

Lightweight, portable relational persistence for Users, Feasibility Report

#### LocalStorage State Sync

Real-Time Continuity

Immediate bi-directional browser state synchronization ensuring zero d

#### 1-Click Demo Evaluator Mode

Instant Access

Pre-configured demo profile enabling one-click test credentials for hassl

# Additional Details(Keep the content limited to this slide only)

## 1. Live Prototype & Verification Highlights

- **1-Click Demo Evaluation Mode:** Pre-configured credentials (demo@gramsaarthi.ai / demo123) allow evaluators instant access to all reports without registration barriers.
- **Tested & Working End-to-End:** Every single user journey—Registration -> Intake Wizard -> Radial Map -> Statutory Calculator -> Trilingual Chat -> Feasibility Dossier -> PDF Export—has been tested and verified with active persistence.
- **Hybrid State Architecture:** Bi-directional sync between browser LocalStorage and Prisma/SQLite database ensures complete data continuity across all route transitions.
- **Print-Optimized Dossier:** Bank-ready Feasibility Report (/report) contains professional print stylesheets that render directly into an institutional PDF loan proposal.

## 2. Alignment with P11 Problem Statement & Screening

- **Exact Problem Statement Alignment:** Strict adherence to P11 rural micro-enterprise advisory guidelines, statutory debt limits, and local market validation.
- **Mathematical Auditability:** Zero floating-point inaccuracies or LLM hallucinated debt figures; statutory equations are completely transparent and verifiable by bank managers.
- **Open Source & Documented:** Comprehensive documentation suite in MD\_Files/ including ARCHITECTURE.md, FEATURES.md, DEMO\_SCRIPT.md, MANUAL\_TEST\_REPORT.md, and TECH\_STACK.md.
- **GitHub Repository:** Publicly accessible codebase with commit history, detailed setup guide, and clean modular Next.js architecture.
- **Demo Video Ready:** Structured 3-minute demonstration script prepared covering problem explanation, live feature walkthrough, and social impact.